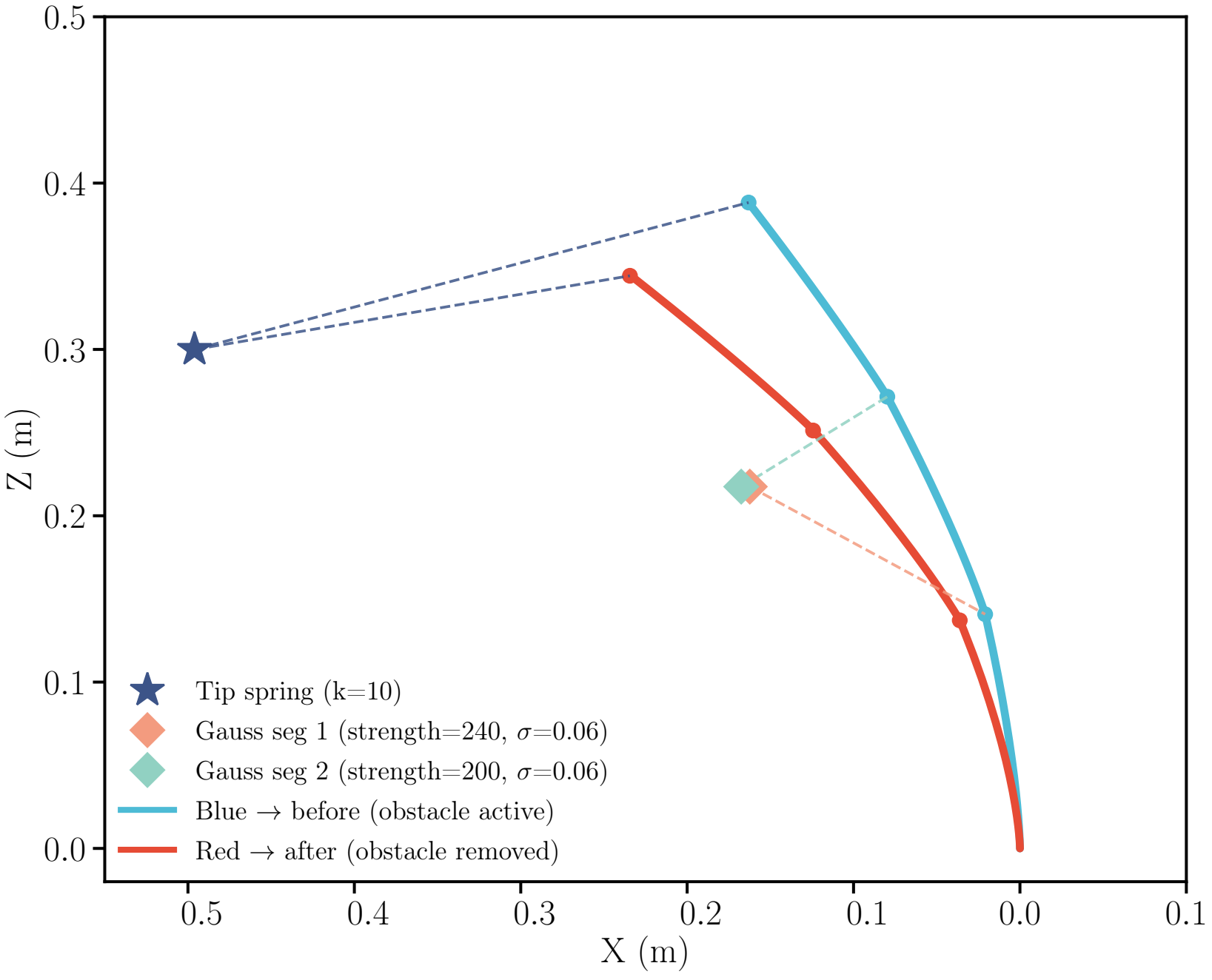

0.5
0.4
0.3
0.2
0.1
0.0
Z (m)
0.5
0.4
0.3
0.2
0.1
0.0
0.1
X (m)
Tip spring (k=10)
Gauss seg 1 (strength=240, σ=0.06)
Gauss seg 2 (strength=200, σ=0.06)
Blue → before (obstacle active)
Red → after (obstacle removed)